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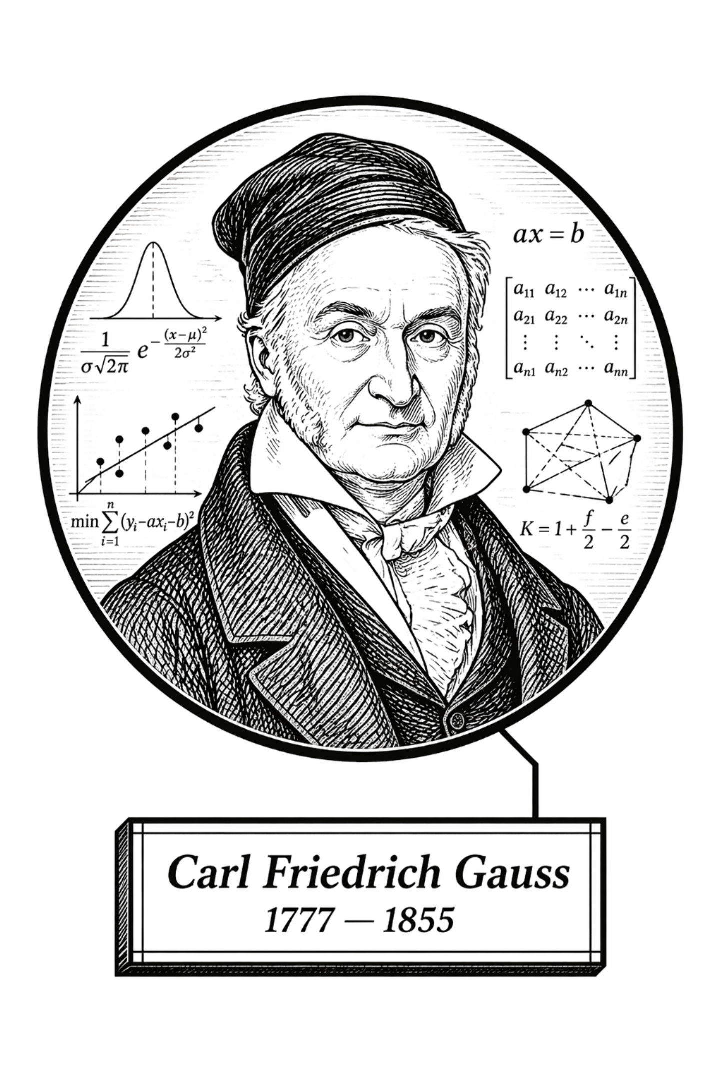

So many squares...

Carl Gauss (probably)

Volume V

Topology and Geometry

Self-Study Notes

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Chapter 1

Euclidean Geometry

Where You Are in the Journey

Logic, Sets, Proof Techniques → **Euclidean Geometry** → Analytical Geometry → Non-Euclidean Geometry → ···

How we got here. Propositional logic and formal proof gave us the tools to reason rigorously. Euclidean geometry is the original domain in which those tools were deployed: Euclid's *Elements* is the first sustained axiomatic system in history, and its influence on the ideal of mathematical proof persists to the present day.

What this chapter builds. Euclid's five postulates and the synthetic method; congruence and similarity of triangles; the Pythagorean theorem and its converses; circle theorems; constructions with compass and straightedge; area and the method of exhaustion; and the logical status of the parallel postulate as a genuinely independent axiom.

Where this leads. Replacing the parallel postulate with its negation yields non-Euclidean geometry. Translating synthetic statements into coordinate equations yields analytical geometry. The independence of the parallel postulate, demonstrated by models of hyperbolic geometry, is also one of the earliest instances of the model-theoretic reasoning developed in Volume I.

Structural Role: Classical Geometry The Axiomatic Template

Euclidean geometry is the historical and logical origin of the axiomatic method. Euclid's program — state your axioms, define your objects, derive everything else — is the template that every other chapter in this project follows. Working through the *Elements* therefore serves a dual purpose: it builds geometric intuition and it makes the axiomatic method visceral and familiar.

The parallel postulate occupies a special place. For two millennia, mathematicians attempted to derive it from the other four. Its independence, established by consistent non-Euclidean models, is one of the great conceptual milestones of mathematics and a direct precursor to the model-theoretic ideas in Volume I.

Structural Roadmap

Definitions $\longrightarrow$ Main Theorems $\longrightarrow$ Consequences and Structural Insight

The global progression is:

1. Euclid's postulates and the synthetic method
2. Congruence criteria: SAS, ASA, SSS
3. The Pythagorean theorem and its converse
4. Similarity and proportionality theorems
5. Circle theorems: inscribed angles, tangents, power of a point
6. Compass-and-straightedge constructions and impossibility results
7. Area via decomposition and the method of exhaustion
8. Hilbert's axioms: a modern repair of Euclid's foundations
9. The logical status of the parallel postulate

Remark (Primary sources). Euclid, *Elements* (Heath translation); Hartshorne, *Geometry: Euclid and Beyond*; Moise, *Elementary Geometry from an Advanced Standpoint*.

Where You Are in the Journey

Euclidean Geometry → **Foundations** → Postulates → Axioms → Synthetic Geometry

How we got here. The chapter-level overview placed Euclidean geometry in the broader development of the axiomatic method. We now begin at the foundation level: the primitive objects, derived notions, and basic structural assumptions from which Euclid's synthetic geometry starts.

What this topic builds. This topic records the foundational vocabulary, postulates, axioms, and propositions used by the Euclidean geometry chapter. Definitions appear first, followed by postulates, axioms, and propositions.

1.1 Foundations

Definitions

The following definitions are Euclid's Book I definitions in the Heath translation [Euclid, 1956, Book I, Definitions 1–23].

Definition 1.1.1 (Point). A point is that which has no part.

Definition 1.1.2 (Line). A line is breadthless length.

Definition 1.1.3 (Ends of a line). The ends of a line are points.

Definition 1.1.4 (Straight line). A straight line is a line which lies evenly with the points on itself.

Definition 1.1.5 (Surface). A surface is that which has length and breadth only.

Definition 1.1.6 (Edges of a surface). The edges of a surface are lines.

Definition 1.1.7 (Plane surface). A plane surface is a surface which lies evenly with the straight lines on itself.

Definition 1.1.8 (Plane angle). A plane angle is the inclination to one another of two lines in a plane which meet one another and do not lie in a straight line.

Definition 1.1.9 (Rectilinear angle). And when the lines containing the angle are straight, the angle is called rectilinear.

Definition 1.1.10 (Right angle and perpendicular). When a straight line standing on a straight line makes the adjacent angles equal to one another, each of the equal angles is right, and the straight line standing on the other is called a perpendicular to that on which it stands.

Definition 1.1.11 (Obtuse angle). An obtuse angle is an angle greater than a right angle.

Definition 1.1.12 (Acute angle). An acute angle is an angle less than a right angle.

Definition 1.1.13 (Boundary). A boundary is that which is an extremity of anything.

Definition 1.1.14 (Figure). A figure is that which is contained by any boundary or boundaries.

Definition 1.1.15 (Circle). A circle is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure equal one another.

Definition 1.1.16 (Center of a circle). And the point is called the center of the circle.

Definition 1.1.17 (Diameter). A diameter of the circle is any straight line drawn through the center and terminated in both directions by the circumference of the circle, and such a straight line also bisects the circle.

Definition 1.1.18 (Semicircle). A semicircle is the figure contained by the diameter and the circumference cut off by it. And the center of the semicircle is the same as that of the circle.

Definition 1.1.19 (Rectilinear figures). Rectilinear figures are those which are contained by straight lines, trilateral figures being those contained by three, quadrilateral those contained by four, and multilateral those contained by more than four straight lines.

Definition 1.1.20 (Equilateral, isosceles, and scalene triangles). Of trilateral figures, an equilateral triangle is that which has its three sides equal, an isosceles triangle that which has two of its sides alone equal, and a scalene triangle that which has its three sides unequal.

Definition 1.1.21 (Right-angled, obtuse-angled, and acute-angled triangles). Further, of trilateral figures, a right-angled triangle is that which has a right angle, an obtuse-angled triangle that which has an obtuse angle, and an acute-angled triangle that which has its three angles acute.

Definition 1.1.22 (Quadrilateral classes). Of quadrilateral figures, a square is that which is both equilateral and right-angled; an oblong that which is right-angled but not equilateral; a rhombus that which is equilateral but not right-angled; and a rhomboid that which has its opposite sides and angles equal to one another but is neither equilateral nor right-angled. And let quadrilaterals other than these be called trapezia.

Definition 1.1.23 (Parallel straight lines). Parallel straight lines are straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.

Postulates

The following are Euclid's five postulates from Book I of the *Elements* in the Heath translation [Euclid, 1956, Book I, Postulates 1–5].

Axiom 1.1.24 (Euclid Postulate I: Drawing a straight line). Let it have been postulated to draw a straight line from any point to any point.

Axiom 1.1.25 (Euclid Postulate II: Producing a finite straight line). Let it have been postulated to produce a finite straight line continuously in a straight line.

Axiom 1.1.26 (Euclid Postulate III: Describing a circle). Let it have been postulated to describe a circle with any center and distance.

Axiom 1.1.27 (Euclid Postulate IV: Equality of right angles). Let it have been postulated that all right angles are equal to one another.

Axiom 1.1.28 (Euclid Postulate V: Parallel postulate). Let it have been postulated that, if a straight line falling on two straight lines makes the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which are the angles less than the two right angles.

Remark (Exposition). Postulates are geometry-specific starting assumptions. They say which constructions are permitted and which basic geometric facts may be used without proof. In Euclid's system, these are not conclusions reached from earlier results; they are part of the foundation from which the theory begins.

Later propositions are proved from the definitions, the postulates, the common notions, and propositions already established. Thus the postulates are the rules of play for Euclidean geometry: they determine what can be constructed what may be assumed about straight lines and circles, and where the theory's geometric content enters before proof begins.

Axioms

Euclid calls the following principles *common notions*. They function as general axioms governing equality and comparison in Book I of the *Elements* [Euclid, 1956, Book I, Common Notions 1–5].

Axiom 1.1.29 (Euclid Common Notion I: Equality is transitive through a common equal). Things which are equal to the same thing are also equal to one another.

Axiom 1.1.30 (Euclid Common Notion II: Adding equals to equals). If equals be added to equals, the wholes are equal.

Axiom 1.1.31 (Euclid Common Notion III: Subtracting equals from equals). If equals be subtracted from equals, the remainders are equal.

Axiom 1.1.32 (Euclid Common Notion IV: Coincidence implies equality). Things which coincide with one another are equal to one another.

Axiom 1.1.33 (Euclid Common Notion V: Whole and part). The whole is greater than the part.

Remark (Exposition). Common notions are general principles of reasoning rather than specifically geometric assumptions. They express basic behavior of equality, comparison, and logical transfer: equal things may be substituted for equal things, adding or subtracting equals preserves equality, and a whole exceeds its part.

Because these principles are not tied to points, lines, circles, or planes, they can be used throughout mathematics, not only in geometry. In modern language, they play the role of background axioms governing how equality and comparison may be used inside proofs.

Where You Are in the Journey

Euclidean Geometry → Foundations → **Compass-and-Straightedge Constructions** → Congruence → Synthetic Geometry

How we got here. The foundations topic fixed Euclid's primitive vocabulary, postulates, and common notions. Those assumptions now become construction rules: straight lines may be drawn, finite straight lines may be produced, and circles may be described from a center and distance.

What this topic builds. This topic begins Euclid's constructive geometry. The first propositions show how the postulates generate objects: an equilateral triangle on a given segment, a copied segment from a given point, and a segment cut off from a larger segment.

1.2 Compass-and-Straightedge Constructions

Propositions

The following construction propositions are Euclid's Propositions I.1–I.3 in modern theorem form, following the Heath translation of the *Elements* [Euclid, 1956, Book I, Propositions 1–3].

Theorem 1.2.1 (Euclid I.1: Constructing an equilateral triangle). *Given a finite straight line segment AB, there exists an equilateral triangle ABC constructed on AB.*

Remark (Proof). Go to proof.

Theorem 1.2.2 (Euclid I.2: Copying a segment from a given point). *Given a point A and a finite straight line segment BC, there exists a straight line segment beginning at A and equal to BC.*

Remark (Proof). Go to proof.

Theorem 1.2.3 (Euclid I.3: Cutting off an equal segment). *Given two unequal finite straight line segments, there exists a construction which cuts off from the greater a straight line segment equal to the less.*

Remark (Proof). Go to proof.

Proofs

Remark (Return). Return to Theorem.

Theorem (Euclid I.1: Constructing an equilateral triangle). *Given a finite straight line segment AB, there exists an equilateral triangle ABC constructed on AB.*

Professional Standard Proof.
TODO: Write the compact canonical construction proof.

Detailed Learning Proof.
TODO: Expand the construction into a detailed learning proof.

Remark (Proof structure). TODO: Record the construction steps and the equality argument.

Remark (Dependencies).

- Point
- Straight line
- Circle
- Equilateral triangle
- Euclid Postulate I
- Euclid Postulate III
- Euclid Common Notion I

Remark (Return). Return to Theorem.

Theorem (Euclid I.2: Copying a segment from a given point). *Given a point A and a finite straight line segment BC, there exists a straight line segment beginning at A and equal to BC.*

Professional Standard Proof.
TODO: Write the compact canonical construction proof.

Detailed Learning Proof.
TODO: Expand the construction into a detailed learning proof.

Remark (Proof structure). TODO: Record the construction steps and the equality transfers.

Remark (Dependencies).

- Point
- Straight line
- Circle
- Euclid I.1
- Euclid Postulate I
- Euclid Postulate II
- Euclid Postulate III
- Euclid Common Notion I

Remark (Return). Return to Theorem.

Theorem (Euclid I.3: Cutting off an equal segment). *Given two unequal finite straight line segments, there exists a construction which cuts off from the greater a straight line segment equal to the less.*

Professional Standard Proof.
TODO: Write the compact canonical construction proof.

Detailed Learning Proof.
TODO: Expand the construction into a detailed learning proof.

Remark (Proof structure). TODO: Record how the copied length and circle determine the cut-off point.

Remark (Dependencies).

- Straight line
- Circle
- Euclid I.2
- Euclid Postulate III
- Euclid Common Notion I

Capstone

To be populated.

Chapter 2

Analytical Geometry

Where You Are in the Journey

Linear Algebra → **Analytical Geometry** → Differential Geometry → Riemannian Geometry → ⋯

How we got here. Linear algebra provided abstract vector spaces and linear maps. Analytical geometry applies this algebraic machinery directly to Euclidean space, giving coordinates, distances, angles, and equations of geometric objects.

What this chapter builds. Coordinates in $\mathbb{R}^n$; dot product, cross product, and orthogonality; lines and planes via parametric and normal-form equations; conic sections and quadric surfaces; projections and reflections; affine maps; and the interplay between geometric intuition and algebraic computation.

Where this leads. Analytical geometry provides the ambient space in which differential geometry operates. The transition from flat Euclidean geometry to curved surfaces begins with the study of curves and surfaces in $\mathbb{R}^3$ and culminates in the intrinsic geometry of manifolds.

Structural Role: Analytical Geometry Algebra Meets Space

Analytical geometry is the translation layer between algebra and space.
Every geometric object a point, line, plane, circle, ellipse has an algebraic description, and every algebraic object with the right structure has a geometric realisation. This duality is the foundation of the rest of the geometry sequence.
The key structural tool is the inner product: it encodes both length and angle in a single algebraic operation. Orthogonality, projection, and reflection all derive from it, and the transition to curved geometry replaces the constant inner product with one that varies from point to point (the metric tensor in Riemannian geometry).

Structural Roadmap

Definitions ⟶ Main Theorems ⟶ Consequences and Structural Insight

The global progression is:

1. Coordinates and the Euclidean inner product
2. Lengths, angles, and the Cauchy–Schwarz inequality
3. Lines and planes: parametric, vector, and scalar forms
4. Projections and orthogonal decompositions
5. Cross product in $\mathbb{R}^3$: definition and geometry
6. Conic sections: circles, ellipses, parabolas, hyperbolas
7. Quadric surfaces: classification via linear algebra

8. Affine maps and rigid motions

Remark (Primary sources). Axler, *Linear Algebra Done Right*, Chapters 6–7; Shafarevich & Remizov, *Linear Algebra and Geometry*; Brannan, Esplen & Gray, *Geometry*.

Status: Planned

Coming Soon

Notes, proofs, and exercises will appear here in a future revision.

Proofs

To be populated.

Capstone

To be populated.

Chapter 3

Trigonometry

trigonometry

Analytical Geometry $\rightarrow$ **Trigonometry** $\rightarrow \cdots$

Proofs

To be populated.

Capstone

To be populated.

Chapter 4

Non-Euclidean Geometry

Where You Are in the Journey

Classical Geometry → Analytical Geometry → **Non-Euclidean Geometry** → Differential Geometry → Riemannian Geometry → ⋯

How we got here. Classical geometry raised the question of the parallel postulate's independence. Analytical geometry gave us the coordinate tools to build explicit models. Non-Euclidean geometry answers the independence question decisively by constructing consistent geometries in which the parallel postulate fails, demonstrating that Euclidean geometry is one possibility among many, not a logical necessity.

What this chapter builds. The Beltrami–Klein and Poincaré disk models of hyperbolic geometry; the failure of the parallel postulate and its consequences for angle sums; elliptic and spherical geometry; isometries of each model; the Gauss–Bonnet theorem in the non-Euclidean setting; and the relationship between non-Euclidean geometries and constant-curvature Riemannian manifolds.

Where this leads. Each non-Euclidean geometry is a special case of a Riemannian manifold with constant sectional curvature: the sphere ($\kappa > 0$), the Euclidean plane ($\kappa = 0$), and hyperbolic space ($\kappa < 0$). These are the model spaces of Riemannian comparison geometry and appear throughout the study of Brownian motion on manifolds.

Structural Role: Non-Euclidean Geometry Geometry is a Choice

Non-Euclidean geometry is the realisation that *geometry is a choice of axioms*, not an inevitable description of space.

The Poincaré disk is a complete, consistent model of hyperbolic geometry living entirely within Euclidean geometry. Its existence proves that the parallel postulate cannot be derived from the other four — if it could, hyperbolic geometry would be inconsistent. This is model-theoretic reasoning in action: independence is established by exhibiting a model.

The curvature interpretation comes later, in Riemannian geometry, where each non-Euclidean geometry is realised as a space of constant sectional curvature. The present chapter builds the geometric intuition; Riemannian geometry provides the analytic framework.

Structural Roadmap

Definitions ⟶ Main Theorems ⟶ Consequences and Structural Insight

The global progression is:

1. The parallel postulate and its historical status

2. Spherical geometry: great circles, angle sums, area

3. The Beltrami–Klein model of hyperbolic geometry
4. The Poincaré disk and upper half-plane models
5. Hyperbolic isometries and Möbius transformations
6. Angle sums and the defect formula in hyperbolic geometry
7. The Gauss–Bonnet theorem in the non-Euclidean setting
8. Elliptic geometry and projective models
9. Constant-curvature spaces as Riemannian manifolds

Remark (Primary sources). Greenberg, *Euclidean and Non-Euclidean Geometries*; Stillwell, *Sources of Hyperbolic Geometry*; Ratcliffe, *Foundations of Hyperbolic Manifolds*.

Status: Planned

Coming Soon

Notes, proofs, and exercises will appear here in a future revision.

Proofs

To be populated.

Capstone

To be populated.

Chapter 5

Differential Geometry

Where You Are in the Journey

Calculus $\to$ Analytical Geometry $\to$ **Differential Geometry** $\to$ Riemannian Geometry $\to$ Stochastic Analysis on Manifolds $\to \cdots$

How we got here. Calculus built the tools of local linear approximation on $\mathbb{R}$. Analytical geometry gave those tools a spatial stage. Differential geometry takes the next step: it studies geometric objects — curves and surfaces — not just as sets of points, but as objects equipped with their own intrinsic notion of calculus.

What this chapter builds. Parametric curves in $\mathbb{R}^3$; arc length, curvature, and the Frenet–Serret frame; regular surfaces in $\mathbb{R}^3$; the first and second fundamental forms; Gaussian and mean curvature; geodesics; and the Gauss–Bonnet theorem as a global structural result connecting curvature to topology.

Where this leads. The notion of a smooth manifold abstracts away the ambient Euclidean space entirely, replacing it with coordinate charts and transition maps. Riemannian geometry equips manifolds with an inner product at each tangent space, recovering notions of length, angle, and curvature in full generality. This is the immediate prerequisite for stochastic analysis on manifolds.

Structural Role: Differential Geometry Calculus on Curved Spaces

Differential geometry is the study of geometric objects through the lens of calculus. The central insight is that curvature — the failure of a space to be flat — can be measured *intrinsically*, without reference to any ambient space. Gauss's Theorema Egregium is the landmark result: Gaussian curvature is preserved under isometries and therefore belongs to the surface itself, not to how it sits in $\mathbb{R}^3$.
The Gauss–Bonnet theorem goes further: it equates the integral of curvature over a compact surface to a purely topological quantity (the Euler characteristic). This is the first glimpse of how local analytic data constrains global topological structure, a theme that runs through all of modern geometry and topology.

Structural Roadmap

Definitions $\longrightarrow$ Main Theorems $\longrightarrow$ Consequences and Structural Insight

The global progression is:

1. Parametric curves: arc length, speed, reparametrisation
2. Curvature and torsion; the Frenet–Serret frame
3. Regular surfaces in $\mathbb{R}^3$: coordinate charts and tangent planes
4. The first fundamental form: lengths and areas on a surface
5. The second fundamental form: normal curvature and shape operator

6. Gaussian and mean curvature; the Theorema Egregium
7. Geodesics: definition, existence, and examples
8. The Gauss–Bonnet theorem
9. Smooth manifolds: charts, atlases, and the passage to intrinsic geometry

Remark (Primary sources). do Carmo, *Differential Geometry of Curves and Surfaces*; Lee, *Introduction to Smooth Manifolds*; Pressley, *Elementary Differential Geometry*.

Status: Planned

Coming Soon

Notes, proofs, and exercises will appear here in a future revision.

Proofs

To be populated.

Capstone

To be populated.

Chapter 6

Riemannian Geometry

Where You Are in the Journey

Differential Geometry $\to$ **Riemannian Geometry** $\to$ Stochastic Analysis on Manifolds $\to$ Brownian Motion on a Torus $\to$ $\cdots$

How we got here. Differential geometry showed how calculus extends to curves and surfaces, and introduced the smooth manifold as the natural domain. Riemannian geometry equips each tangent space of a manifold with an inner product that varies smoothly from point to point — the Riemannian metric. This single addition restores all of the metric concepts (distance, angle, volume, curvature) in the fully intrinsic setting.

What this chapter builds. Riemannian metrics and the metric tensor; the Levi-Civita connection; geodesics as length-minimising curves and via the geodesic equation; parallel transport; the Riemann curvature tensor; sectional, Ricci, and scalar curvature; the exponential map; and comparison geometry.

Where this leads. The Levi-Civita connection is the intrinsic derivative that stochastic analysis on manifolds requires. Hsu's *Stochastic Analysis on Manifolds* builds Brownian motion directly on Riemannian manifolds using the connection, the exponential map, and the Laplace–Beltrami operator — all developed in this chapter.

Structural Role: Riemannian Geometry Intrinsic Metric Structure

Riemannian geometry is the culmination of the geometry sequence and the direct prerequisite for the project's terminal goal: stochastic analysis on manifolds.

The Riemannian metric is the manifold analogue of the inner product. It makes precise what it means for two tangent vectors to be perpendicular, for a curve to have a definite length, and for a region to have a definite volume — all without any reference to an ambient space.

The Levi-Civita connection is the unique connection compatible with the metric and torsion-free. It defines parallel transport, covariant derivatives, and the geodesic equation. These are the tools that replace ordinary derivatives when working on curved spaces, and they are exactly what is needed to define stochastic differential equations on manifolds.

The Laplace–Beltrami operator, the Riemannian generalisation of the ordinary Laplacian, is the infinitesimal generator of Brownian motion on a Riemannian manifold.

Structural Roadmap

Definitions $\longrightarrow$ Main Theorems $\longrightarrow$ Consequences and Structural Insight

The global progression is:

1. Riemannian metrics: definition, examples, and the metric tensor

2. Lengths of curves and the Riemannian distance function
3. Connections and covariant derivatives
4. The Levi-Civita connection: existence and uniqueness
5. Parallel transport along curves
6. Geodesics: the geodesic equation and the exponential map
7. The Riemann curvature tensor: definition and symmetries
8. Sectional curvature, Ricci curvature, and scalar curvature
9. The Laplace–Beltrami operator
10. Model spaces: spheres, hyperbolic space, flat tori
11. Comparison geometry and the Hopf–Rinow theorem

Remark (Primary sources). Lee, *Riemannian Manifolds: An Introduction to Curvature*; do Carmo, *Riemannian Geometry*; Hsu, *Stochastic Analysis on Manifolds*, Chapter 1 (for the stochastic-analysis perspective on the Riemannian toolkit).

Status: Planned

Coming Soon

Notes, proofs, and exercises will appear here in a future revision.

Proofs

To be populated.

Capstone

To be populated.

Chapter 7

Topology

Status: Planned

Active mathematical content has not yet been added.

Topology Notes

This chapter is planned. Active mathematical content has not yet been added.

Proofs

Proof entries for this chapter are planned. No proof files have been regenerated.

Exercises

Exercises for this chapter are planned. No exercise content has been restored here yet.

Bibliography

Euclid. *The Thirteen Books of Euclid's Elements*. Dover Publications, New York, second edition, 1956. Translated with introduction and commentary by Thomas L. Heath.